Title

AI Resume Screening System - Project Synopsis

Objective

Build a machine learning application that classifies uploaded resumes into job categories using TF-IDF vectorization and a Naive Bayes classifier.

Dataset

A CSV dataset containing resume text and corresponding job category labels.

Technologies

Python, Pandas, Scikit-learn, Streamlit, Joblib/Pickle.

Workflow

Load dataset, preprocess text, convert text to TF-IDF features, encode labels, split data, train Multinomial Naive Bayes model, save model/vectorizer, build Streamlit interface to upload a PDF resume, extract text, transform with TF-IDF and predict the category.

Modules

pandas, sklearn.feature_extraction.text.TfidfVectorizer, sklearn.preprocessing.LabelEncoder, sklearn.model_selection.train_test_split, sklearn.naive_bayes.MultinomialNB, streamlit, pdfplumber or PyPDF2 for PDF extraction.

Outcome

Users upload a resume PDF and the application predicts the most suitable job category based on the trained model.

Future Scope

Support multiple resume formats, ranking against job descriptions, skill extraction, ATS scoring, and recommendation engine.